MD OMOR FARUK RAHAT [CURRICULUM VITAE]



Software Engineering | Undergraduate

Address: Room 1813, Block B, Wanhe Mansion, High-Tech Zone, Zhengzhou, Henan, China (450001)

Contact: (+86) 16696171024 | **Email:** rahatmdomorfaruk@gmail.com | **WeChat:** omorfarukra

Nationality: Bangladeshi | **Date of Birth:** 26/03/2004

Profiles: GitHub: github.com/omor-rahat | LinkedIn: linkedin.com/in/omor-faruk-rahat

RESEARCH INTERESTS & ACADEMIC OBJECTIVE

Software Engineering undergraduate at Zhengzhou University (National Double First-Class University) with an officially certified an **82/100** cumulative average across 108.0 credits, and an accelerated **3.74 Semester GPA** in Year 3. Officially certified in **HSK Level 4** and holding University **Medium of Instruction (MOI) in English**, offering zero cultural/linguistic friction and immediate research productivity in China. Actively seeking a Master's degree opportunity under the Chinese Government Scholarship (CSC) or University High-Level Postgraduate Schemes for Autumn 2027 to conduct graduate research in **Artificial Intelligence, Natural Language Processing (NLP), High-Performance Computing, Computer Systems & Networks, and Cybersecurity**.

EDUCATION & ACADEMIC BACKGROUND

Zhengzhou University — **Zhengzhou, Henan, China**

School of International Education | B.Eng. in Software Engineering (4-Year Full-Time) — *Sept, 2023 – July, 2027 (Expected)*

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- **Academic Standing:** Cumulative GPA: **3.08 / 4.00** | Overall Weighted Average: **82/100** across 108.0 completed credits.
- **Medium of Instruction (MOI):** Formally certified English Medium of Instruction by Zhengzhou University confirming all undergraduate engineering curriculum instructed in English.
- **Academic Momentum:** Demonstrated strong upward academic growth across upper-division semesters, advancing to a **3.74 Semester GPA** in Year 3 coursework.

Alamgir Monsur (Mintu) Memorial College — **Mymensingh, Bangladesh**

Higher Secondary Certificate (HSC) in Science — *Sept 2020 – Feb 2023*

OFFICIAL CERTIFICATIONS & LANGUAGE PROFICIENCY

- **HSK Level 4 Official Certificate (Chinese Proficiency Test):** Certified by Center for Language Education and Cooperation (CLEC) / Ministry of Education, China. Completed 4 consecutive semesters of university-level Chinese (HSK-2, HSK-3, HSK-4 Vol. 1, HSK-4 Vol. 2). Fully adapted to daily campus life, technical discussions, and academic collaboration in Chinese.
- **Medium of Instruction (MOI) Certificate in English:** Officially certified by Zhengzhou University confirming full 4-year degree instruction in English.
- **English Language Proficiency:** Full Professional & Academic Working Proficiency (CEFR B2 across Listening, Reading, Writing, and Speaking).
- **Bengali:** Native language proficiency.

CORE ACADEMIC MASTERY & HIGHEST-SCORING SUBJECT PORTFOLIO

Certified official transcript coursework evaluated under 4.0 Grade Point system at Zhengzhou University, highlighting elite performance in systems, intelligent algorithms, and engineering fundamentals:

Systems, Networks & Hardware	Grade / GP	AI, Mathematics & Software	Grade / GP
College Physics A (I)	**97 / 100 (4.0)**	Mobile Programming (Android)	**92 / 100 (4.0)**
Computer Organization & Architecture	**91 / 100 (4.0)**	Object-Oriented Java Programming	**91 / 100 (4.0)**
Computer Networks	**90 / 100 (4.0)**	Introduction to Artificial Intelligence	**89 / 100 (3.7)**
High-Performance Computing (Parallel)	**86 / 100 (3.7)**	Discrete Mathematics	**88 / 100 (3.7)**
Introduction to Embedded Systems	**85 / 100 (3.7)**	C Programming Language	**87 / 100 (3.7)**
College Physics Experiments A (I & II)	**87 & 86 (3.7)**	Calculus A (I & II)	**82 & 81 (3.2)**

RESEARCH & TECHNICAL PROJECTS PORTFOLIO

Smart Energy Meter Simulation & Embedded Telemetry System — *Nov 2025*

Embedded Systems Lead | Tech: STM32F103C8T6 (ARM Cortex-M3), Embedded C/C++, Keil uVision, Proteus, LaTeX

- Modeled an IoT-ready microcontroller firmware system capable of continuous 12-bit ADC voltage and current sampling, digital filtering, and real-time active power computation.
- Programmed hardware interrupt-driven surge detection routines with automated emergency circuit tripping and serial UART telemetry transmission for automated energy reporting.
- Authored publication-grade engineering specifications, register configurations, and mathematical validation models formatted in LaTeX.

Auto File Organizer – Intelligent Desktop Automation Utility *June 2025*

Python Developer | Tech: Python 3, Tkinter GUI, Multithreading, OOP Design Patterns

- Engineered an automated desktop application for rapid rule-based file categorization, extension mapping, directory batch indexing, and regex-based custom rule sorting.
- Employed worker threading to maintain non-blocking UI response during heavy file operations; integrated file collision resolution and robust transaction error recovery mechanisms.

Campus Pre-Owned Market System (Software Architecture & Requirements) *Sept 2025 – Jan 2026*

Systems Modeler & Requirements Analyst | Methodologies: IEEE 830 SRS, PlantUML, UML 2.0, Decision Trees

- Formulated complete Software Requirement Specifications (SRS) for a university campus trading ecosystem featuring user verification, secure peer-to-peer escrow, and dispute arbitration pipelines.
- Rendered detailed PlantUML behavioral models (sequence flows, activity diagrams, state charts) and formal truth tables to verify edge-case transactional integrity.

Android Mobile Application Development Laboratory Suite *Oct – Dec 2025*

Mobile Application Developer | Tech: Java, Android Studio, Android SDK, SQLite, XML, Background Services

- Programmed 7 comprehensive laboratory applications exploring lifecycle handling, SQLite relational storage, custom adapters, hardware sensors, and RESTful API consumption.
- Awarded a cohort-leading score of **92 / 100 (4.0 GP)** in Mobile Programming for modular codebase structure and UI/UX efficiency.

Foundational Machine Learning Algorithms & Neural Network Exploration *July 2026*

Applied AI Researcher | Tech: Python 3, NumPy, Pandas, Scikit-Learn, PyTorch, Matplotlib, Jupyter Notebooks

- Executed an intensive research program implementing core supervised and unsupervised algorithms (linear/logistic regression, SVM, decision trees, k-means) and gradient descent optimizers from scratch.
- Prototyped multi-layer perceptrons (MLP) and basic CNN architectures in PyTorch for image classification benchmarks, establishing empirical readiness for graduate NLP and AI research.

TECHNICAL & SCIENTIFIC SKILLS

- **Programming Languages:** Python, Java (SE & Android SDK), C, SQL.
- **AI & Machine Learning:** PyTorch, Scikit-Learn, NumPy, Pandas, Matplotlib, Data Preprocessing, Gradient Descent, Model Evaluation.
- **Systems & Hardware Platforms:** Linux/Unix Shell, Android Studio, STM32 Microcontrollers (ARM Cortex-M3), Keil uVision, Proteus, Git / GitHub.
- **Methodologies & Software Engineering:** Object-Oriented Programming (OOP), IEEE 830 SRS, Software Architecture, UML Modeling, Multithreading, Relational Database Design.
- **Academic Documentation & Productivity:** LaTeX & Overleaf (publication-grade typesetting), PlantUML, Markdown, NotebookLM, Canva, Microsoft Office Suite.

HONORS, ACADEMIC STANDING & OFFICIAL CREDENTIALS

- **Official Pre-Graduation Verification:** Formally certified undergraduate standing with expected degree conferral on **July, 2027** by Zhengzhou University School of International Education.
- **Medium of Instruction (MOI) Certificate:** Formally certified English Medium of Instruction by Zhengzhou University.
- **Official HSK-4 Credential:** National Chinese Proficiency Level 4 certified, confirming language competence for study and research in China.
- **Academic Momentum Recognition:** Attained an exceptional term GPA of **3.74** in upper-division systems coursework, demonstrating sustained scientific dedication.

ACADEMIC REFEREES & RECOMMENDERS

Official recommendation letters and academic verifications are available upon request from the following faculty members at Zhengzhou University:

Referee 1 (Systems & Mobile)	Referee 2 (Networks & HPC)	Referee 3 (AI & Embedded)
Prof. / Dr. [Instructor Name 1]	**Prof. / Dr. [Instructor Name 2]**	**Prof. / Dr. [Instructor Name 3]**
Associate Professor / Course Lead	Professor / Academic Advisor	Associate Professor / Lab Advisor
School of Software / SIE	School of Computer & AI	Dept. of Software Engineering
Zhengzhou University (ZZU)	Zhengzhou University (ZZU)	Zhengzhou University (ZZU)
Focus: Mobile Dev (92) & Java (91)	*Focus: Networks (90) & HPC (86)*	*Focus: AI (89) & Embedded (85)*
Email: `[prof1]@zzu.edu.cn`	Email: `[prof2]@zzu.edu.cn`	Email: `[prof3]@zzu.edu.cn`
Phone: +86-371-xxxxxxxx	Phone: +86-371-xxxxxxxx	Phone: +86-371-xxxxxxxx